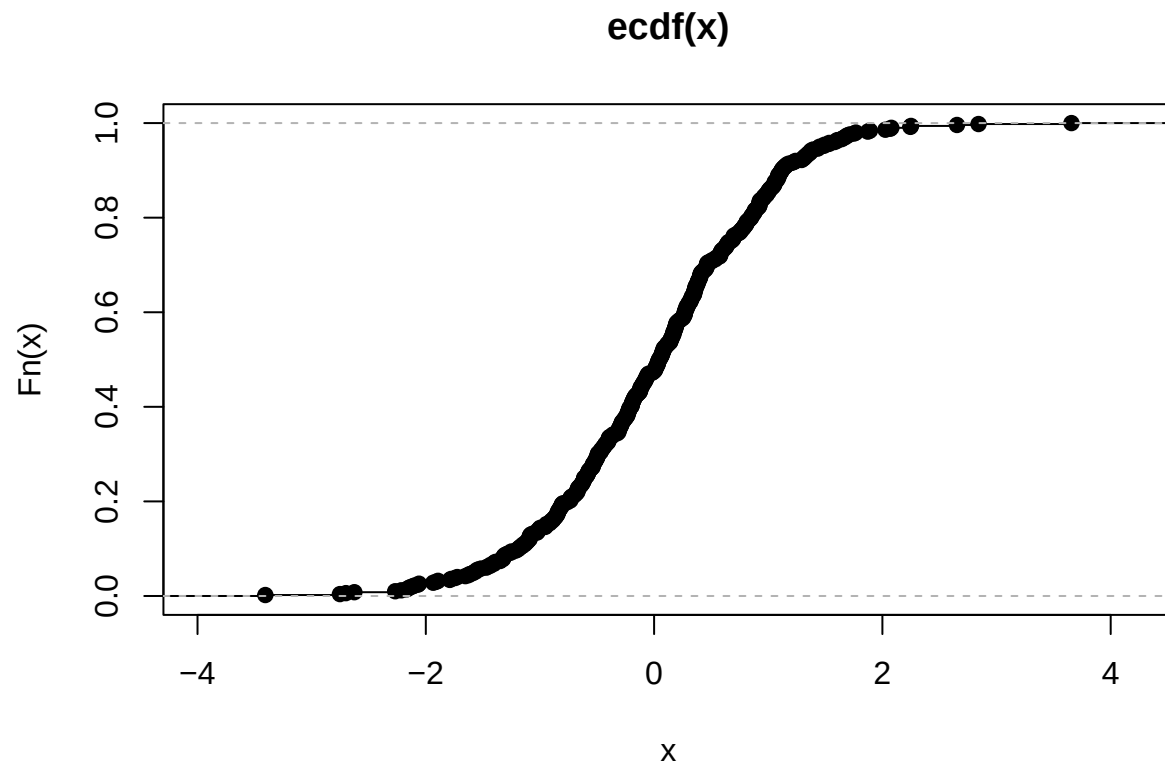


x— title: “week5-1” author: “Greg T.” date: “2026-09-22” output: html_document —

```
# Make an ecdf and plot it:
```

```
x <- rnorm(500,0,1)
Fhat <- ecdf(x)
plot(Fhat)
```



R Markdown

In class work for Week-5

```
n_values <- c(1, 5, 20, 100)
B <- 10000

colors <- c("blue", "green", "orange", "purple")

# Set up an empty plot
plot(
  NA,
  xlim = c(-4, 4),
  ylim = c(0, 1),
  xlab = "x",
  ylab = "CDF",
```

```

    main = "Convergence in Distribution to N(0,1)"
  )

  for (i in 1:length(n_values)) {

    n <- n_values[i]

    x <- rnorm(
      B,
      mean = 0,
      sd = sqrt(1 + 1/n)
    )

    F_hat <- ecdf(x)

    curve(
      F_hat(x),
      from = -4,
      to = 4,
      add = TRUE,
      col = colors[i],
      lwd = 2
    )
  }

  # Limiting theoretical CDF
  curve(
    pnorm(x),
    from = -4,
    to = 4,
    add = TRUE,
    col = "black",
    lwd = 3,
    lty = 2
  )

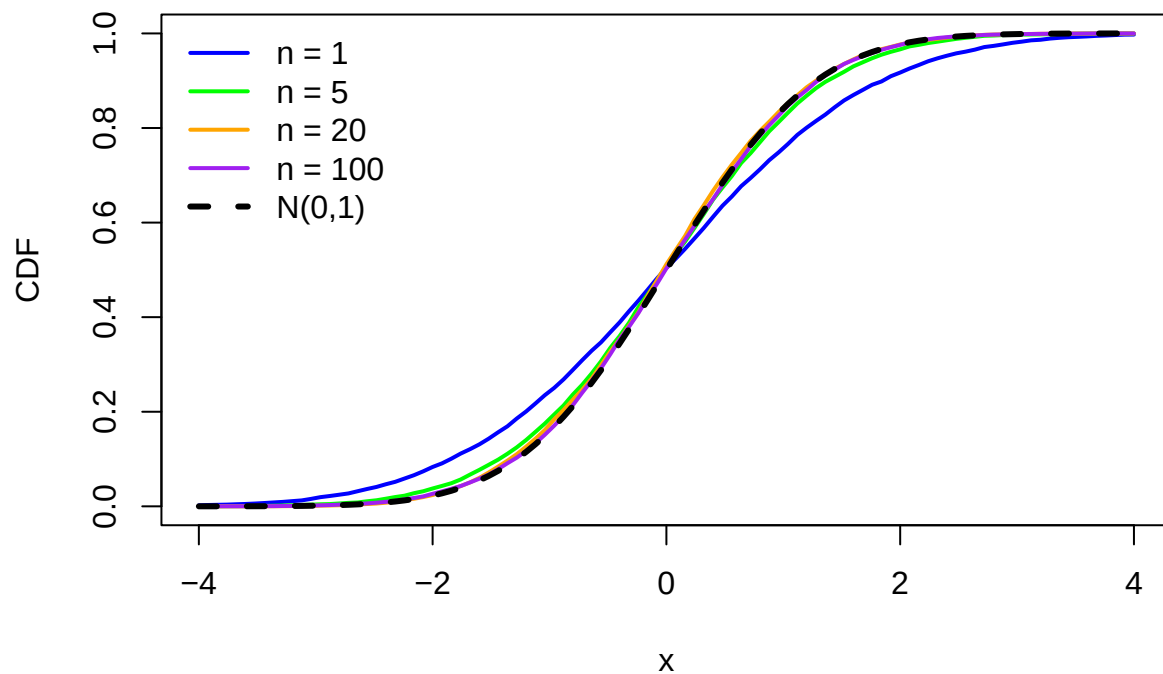
  legend(
    "topleft",
    legend = c(
      paste("n =", n_values),
      "N(0,1)"
    ),
    col = c(colors, "black"),
    lwd = c(rep(2, 4), 3),
    lty = c(rep(1, 4), 2),
    bty = "n"
  )

  mtext(
    "Convergence in Distribution to N(0,1)",
    outer = TRUE,
    cex = 1.3
  )

```

)

Convergence in Distribution to $N(0,1)$



```
# Sim for Bernoulli RV, CLT demonstration

#sample_size <- c(5, 10, 20, 50, 100, 200, 500, 1000)
sample_size <- c(5, 50, 200, 1000)
p = 0.3

par(
  mfrow = c(2, 2),
  oma = c(0, 0, 3, 0)
)

sim_size <- 10000
mean_vector <- numeric(sim_size)

for (i in 1:length(sample_size)){

  n = sample_size[i]

  for (j in 1:sim_size){
    x = rbinom(n, size=1, prob = p)
    x_bar = mean(x)
    mean_vector[j] <- x_bar
  }
}
```

```

}

hist(
  mean_vector,
  probability = TRUE,
  main = paste("n =", n),
  xlab = "x",
  xlim = c(0, 1),
  ylab = "Density"
)

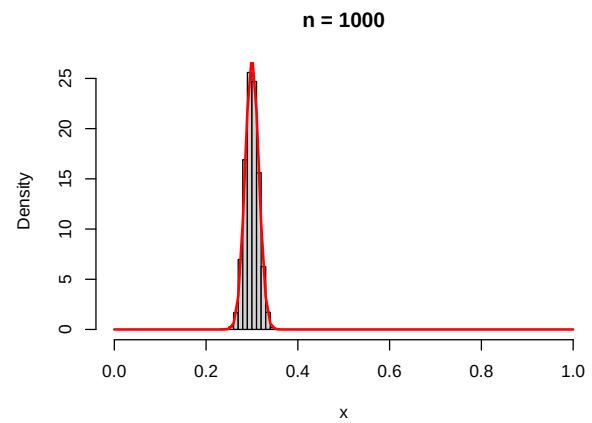
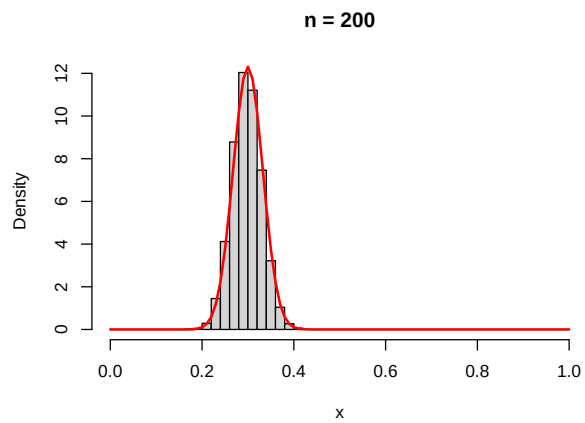
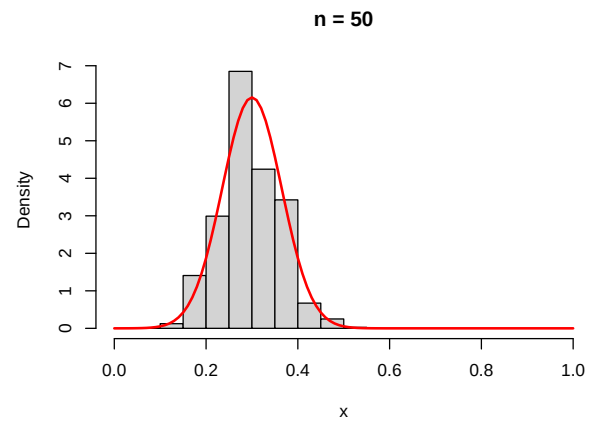
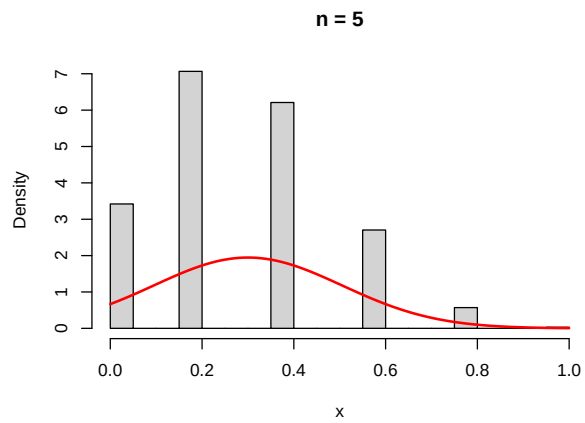
curve(
  dnorm(x, mean = p, sd = sqrt( p*(1-p) /n )),
  add = TRUE,
  col = "red",
  lwd = 2
)

}

mtext(
  "CLT Simulation for Bernoulli, p=0.5",
  outer = TRUE,
  side = 3,
  line = 1,
  cex = 1.5,
  font = 2
)

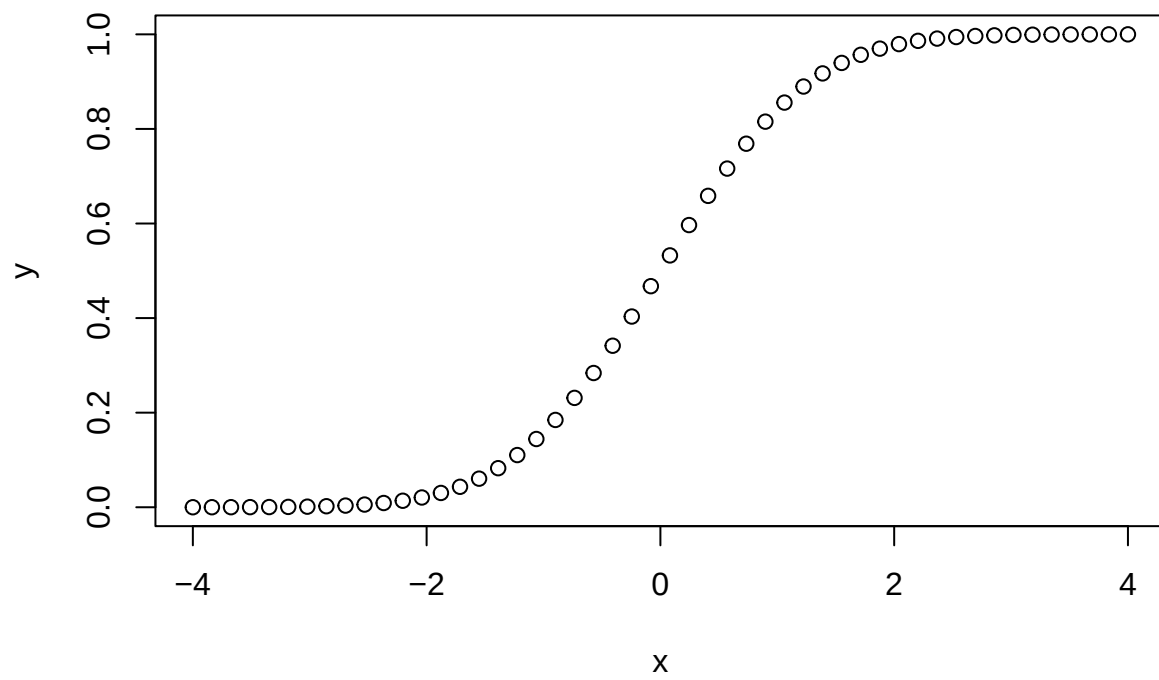
```

CLT Simulation for Bernoulli, $p=0.5$



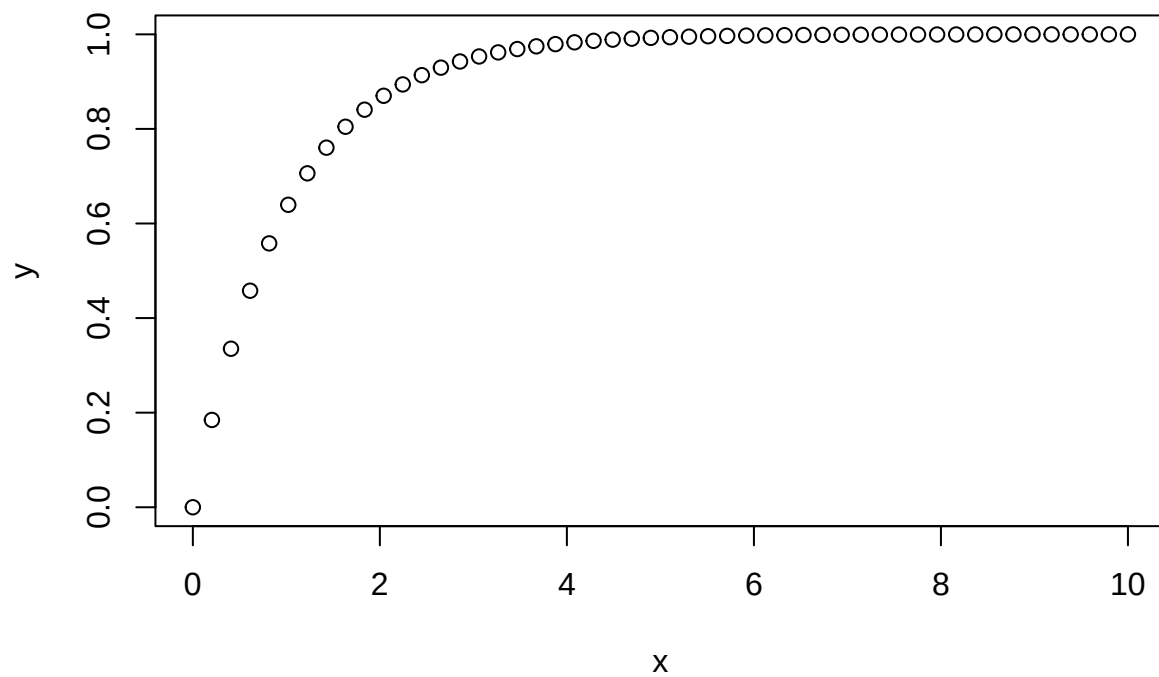
```
n=10  
  
x<- seq(from=-4, to=4, length=50)  
  
y<- pnorm(x)  
  
plot(x,  
      y,  
      main = "CDF of Normal"  
  
      )
```

CDF of Normal



```
n=10  
  
x<- seq(from=0, to=10, length=50)  
  
y<- pexp(x)  
  
plot(x,  
      y,  
      main = "CDF of Exponential"  
  
)
```

CDF of Exponential



```
# CLT Simulaton for Exponential Distribution
```

```
n <- 100
```

```
p <- 0.5
```

```
x <- 0:n
```

```
y <- pbinom(x, size = n, prob = p)
```

```
plot(
```

```
  x,
```

```
  y,
```

```
  type = "s",
```

```
  main = "CDF of Binomial",
```

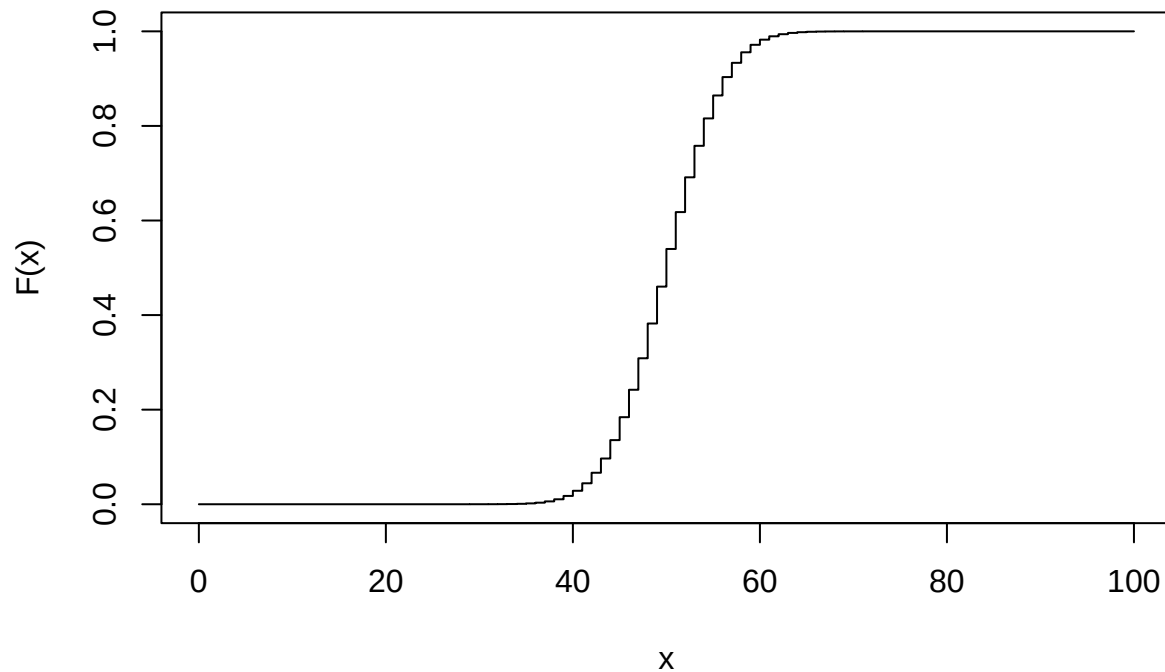
```
  xlab = "x",
```

```
  ylab = "F(x)",
```

```
  ylim = c(0, 1)
```

```
)
```

CDF of Binomial



```
#sample_size <- c(5, 10, 20, 50, 100, 200, 500, 1000)
sample_size <- c(5, 50, 200, 1000)

par(
  mfrow = c(2, 2),
  oma = c(0, 0, 3, 0)
)

sim_size <- 10000 # controls the accuracy of the histogram
                  # The larger the sim size, the closer the sim will get
                  # to the actual pdf

mean_vector <- numeric(sim_size)

for (i in 1:length(sample_size)){

  n = sample_size[i]

  for (j in 1:sim_size){
    x = rexp(n=n, rate = 1/5)
    x_bar = mean(x)
    mean_vector[j] <- x_bar
  }
}
```

```

hist(
  mean_vector,
  probability = TRUE,
  main = paste("n =", n),
  xlab = "x",
  xlim = c(0, 10),
  ylab = "Density"
)

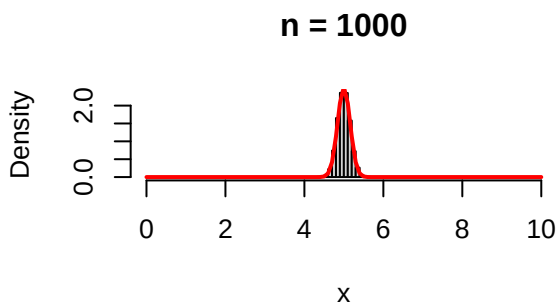
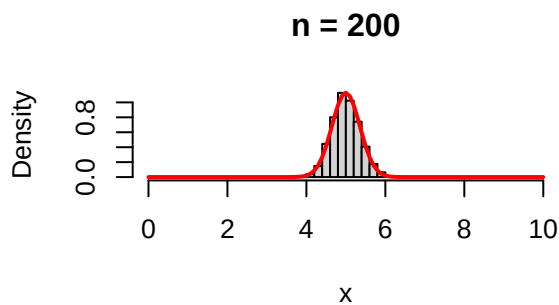
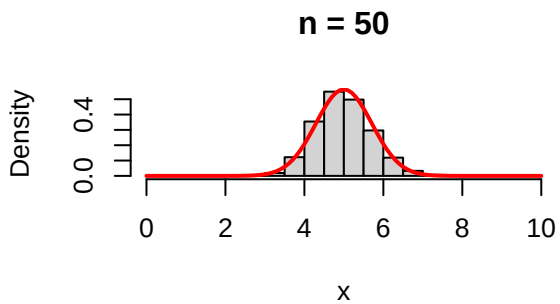
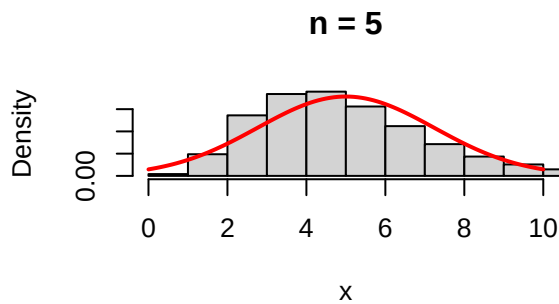
curve(
  dnorm(x, mean = 5, sd = 5 / sqrt(n)),
  add = TRUE,
  col = "red",
  lwd = 2
)

}

mtext(
  "CLT Simulation for exponential dist. ",
  outer = TRUE,
  side = 3,
  line = 1,
  cex = 1.5,
  font = 2
)

```

CLT Simulation for exponential dist.



```
# Histogram approximation of an exponential distribution.

x <- rexp(500, rate = 1/5)

hist(x,
     breaks = 50,
     probability = TRUE,
     main = "Approximation of Exponential Distribution lambda = 5 "
)

curve(
  dexp(x, rate = 1/5),
  add = TRUE,
  col = "RED",
  lwd = 2
)
```

Approximation of Exponential Distribution $\lambda = 5$

